

TCV- ϕ VIII: toward a one-scalar completion from cohesive-vacuum topology and coherent-sector Φ_2 emergence

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This paper investigates whether the ultralight Φ_2 sector of the canonical TCV- ϕ programme can be reinterpreted as an emergent collective mode of the CC- Φ_1 cohesive microstructure. The study is organized as a reproducible inverse/direct pipeline: PMNS-like flavour constraints are first translated into modal and topological requirements, candidate cluster geometries are then compared under common diagnostics, and the leading candidates are finally tested against Φ_2 emergence, reduced homogeneous ULDM-like evolution, and microscopic closure criteria. The main result is structural rather than theorem-level: closed twisted topologies are consistently favored over rigid or open cluster toys, and the twisted-multi-ring geometry gives the strongest combined PMNS + Φ_2 bridge signal. For the full twisted-multi-ring bridge we obtain a calibrated emergent mass $m_2 \simeq 8.98 \times 10^{-23}$ eV, joint and robustness pass fractions close to unity and 0.72 respectively, and a reduced homogeneous evolution compatible with $\rho \propto a^{-3}$. Dedicated micro-closure tests remain partial at the operator level, but residual diagnostics show that the failure is dominated by omitted higher internal modes rather than by inter-ring topology itself. When the reduction is restricted to a coherent global sector, the effective closure improves significantly, which motivates the interpretation of Φ_2 as a coherent-sector effective mode of CC- Φ_1 rather than as a full microscopic closure of the entire microstructure. The present status is therefore a structured one-field route with strong topology-sensitive support and a sector-restricted effective closure, while strict global micro-closure remains open.

I. INTRODUCTION

Papers I–VII established the canonical two-scalar EFT layer of TCV- ϕ and its main phenomenological sectors: background evolution, perturbations, compact objects, and Planck confrontation. The remaining conceptual question is whether Φ_2 must remain a separate effective scalar, or whether it can instead emerge as a collective low-energy mode of the CC- Φ_1 cohesive microstructure.

This paper is written in a deliberately conservative spirit. It does not claim a final microscopic proof. Instead, it organizes the one-field question into a reproducible programme with explicit diagnostics, stop rules, and control geometries. Its status is that of a structured one-field benchmark paper rather than a final closure theorem.

II. SCOPE AND METHODOLOGICAL STATUS

The workflow adopted here has four stages:

- 1) infer modal and geometric requirements from PMNS-like flavour structure;
- 2) compare geometry families under common PMNS and structure scores;
- 3) test whether the same leading geometries support an emergent Φ_2 mode;
- 4) evaluate partial microscopic closure and one-field consistency.

All results quoted below are script-backed and archived in the Paper-08 workspace. We distinguish throughout between:

- exact linear-algebra diagnostics;
- heuristic effective-field reductions;
- exploratory micro-closure attempts.

When a result is obtained from a simplified bridge or family-level screening stage, it is explicitly labelled as such; the strongest claims in this paper are reserved for the dedicated twisted-multi-ring PMNS, full-bridge, residual-diagnostic, and dedicated micro-closure layers.

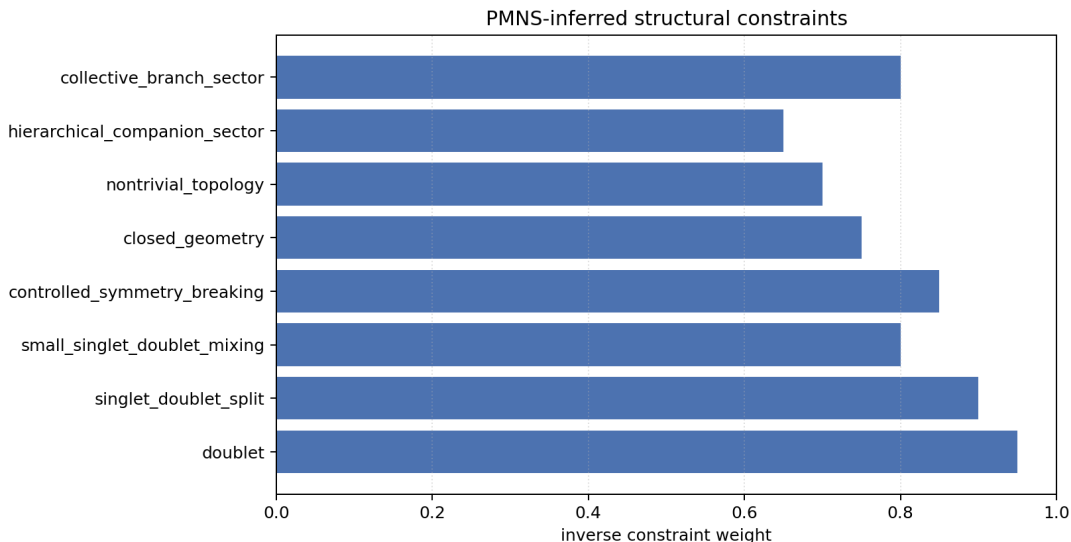


FIG. 1. Inverse PMNS constraint profile used to guide the geometry scan.

Candidate	S_{PMNS}	Strict fraction within natural scan
Twisted torus	0.519	0.0194
Twisted multi ring	0.235	0.0415

TABLE I. Dedicated PMNS-level comparison within the leading twisted-topology family.

III. INVERSE/DIRECT GEOMETRY PROGRAMME

The first step is an inverse PMNS analysis: starting from large θ_{23} , small but non-zero θ_{13} , and moderate θ_{12} , we infer a preferred modal structure with

- a quasi-degenerate doublet,
- a singlet+doublet split,
- weak singlet-doublet mixing,
- controlled symmetry breaking,
- and a preference for closed non-trivial topology.

These inferred constraints are then used to rank explicit geometry families.

The resulting family-level rankings show that rigid geometries remain disfavored, while twisted closed topologies dominate both PMNS and bridge-oriented structural scores. This is the point where geometry becomes a dynamical ingredient rather than a decorative model choice. At this stage, however, the rankings should still be read as family-selection diagnostics rather than as candidate-level proofs.

IV. FROM TWISTED TORUS TO TWISTED MULTI RING

The historical turning point of the programme was the twisted-torus cluster. It was the first explicit geometry to produce a coherent joint signal for PMNS-like flavour and emergent-mode diagnostics. However, the same systematic scan that validated the topological direction also suggested that the twisted torus is not the endpoint of that family.

Two candidate extensions were tested at the same broad topological level: phase-loop and twisted-multi-ring. The phase-loop proxy looked promising but failed to maintain its performance when promoted to a dedicated flavour toy. By contrast, twisted-multi-ring remained strong under dedicated PMNS tests and therefore became the main candidate.

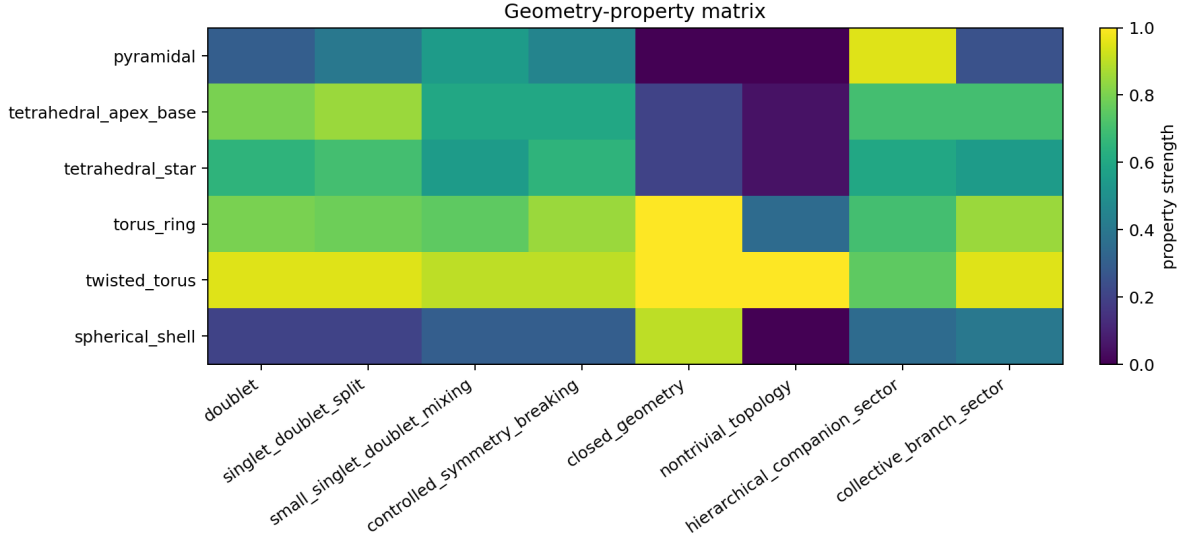


FIG. 2. Geometry-property matrix for the inverse/direct scan. Closed twisted topologies satisfy the largest subset of inferred constraints.

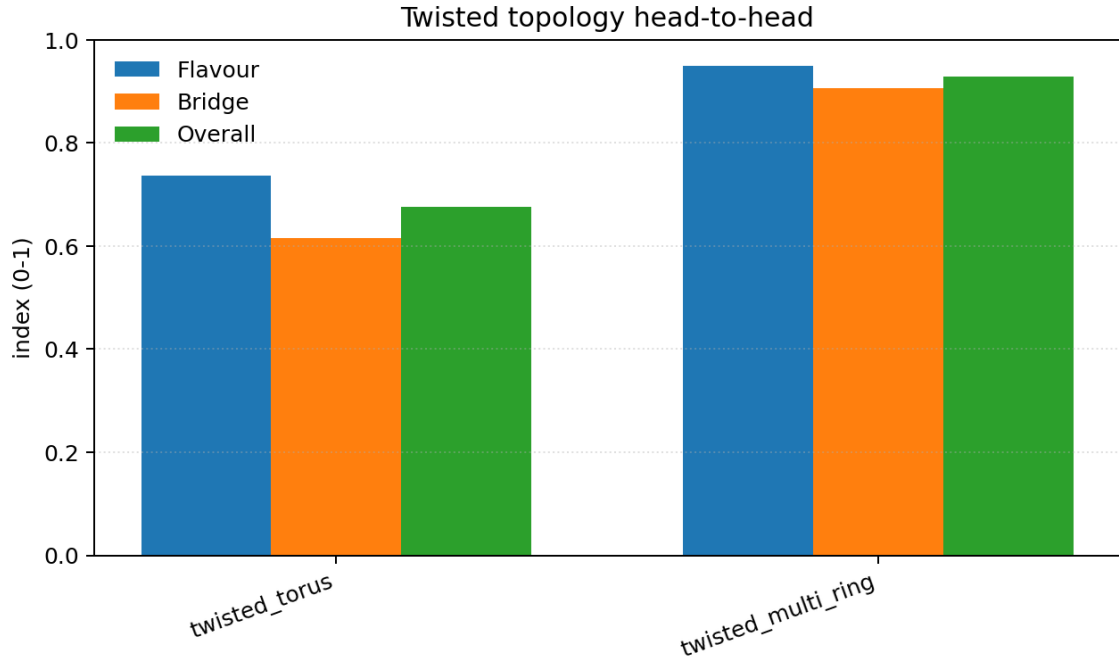


FIG. 3. Head-to-head comparison between the historical twisted-torus benchmark and the current twisted-multi-ring candidate.

The gain is not limited to a single benchmark point. For the twisted-multi-ring sector-compatibility run, the strict lepton robustness fraction reaches

$$f_{\text{strict}}^{(\ell)} \simeq 0.874, \quad (1)$$

while the quark-like toy remains weakly mixed. This supports the interpretation that twisted-multi-ring is not merely a numerically lucky refinement, but a stronger geometry in the same closed twisted family. The role of twisted torus in this paper is therefore historical and methodological, while the role of twisted-multi-ring is candidate-level and prospective.

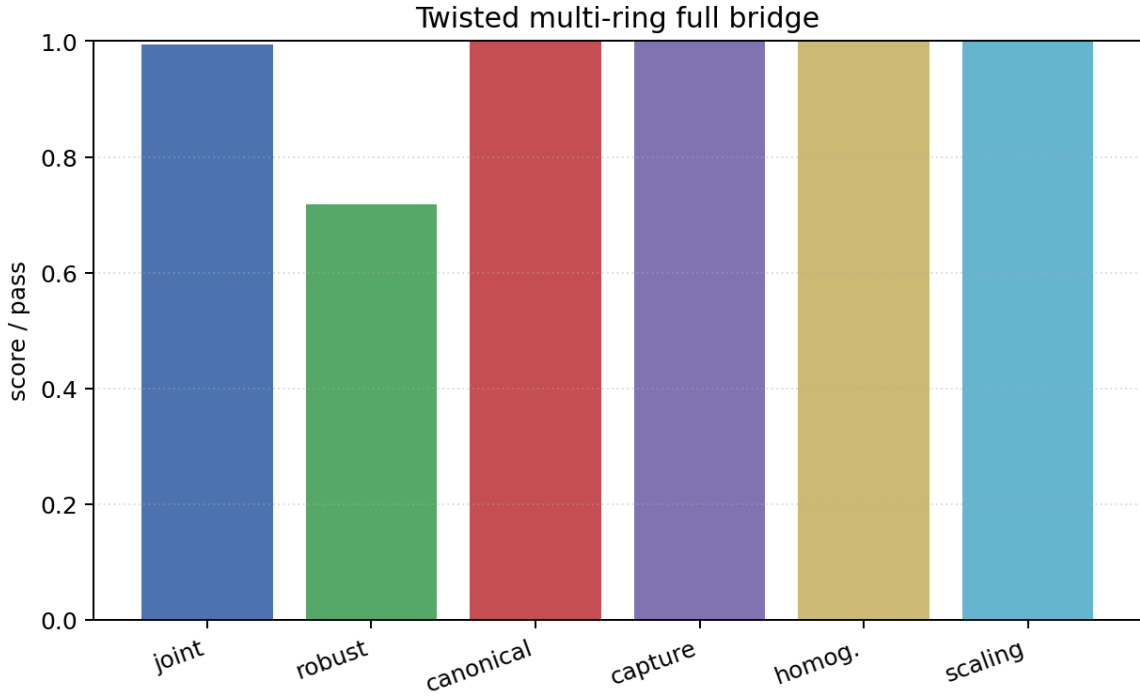


FIG. 4. Full twisted-multi-ring bridge diagnostics.

V. EMERGENT Φ_2 FROM THE TWISTED TOPOLOGY FAMILY

The central question is whether the same geometry that improves flavour also supports a viable emergent Φ_2 mode. For this reason, the leading geometries are passed through a dedicated emergence pipeline evaluating:

- light-mode softness and collectiveness,
- effective mass calibration,
- canonical capture and topology-aware source overlap,
- reduced homogeneous ULDM-like evolution,
- and size scaling.

In the present bridge layer, the quoted physical mass is obtained from a calibrated proxy. More precisely, the dimensionless light-mode quantity $m_{2,\text{proxy}}$ is mapped to physical units through the same reference normalization used in the family-level bridge stage, with a ULDM anchor scale of 10^{-22} eV. The relevant quantity for interpretation is therefore not only the absolute calibrated value, but also the logarithmic deviation from the canonical ULDM window.

At the family level, the twisted-torus benchmark already showed that topology matters. At the dedicated-candidate level, twisted-multi-ring is stronger.

For the full twisted-multi-ring bridge we obtain

$$f_{\text{joint}} \simeq 0.994, \quad f_{\text{robust}} \simeq 0.718, \quad m_2 \simeq 8.98 \times 10^{-23} \text{ eV}, \quad (2)$$

with all current bridge checks passing, including canonical normalization, topology-aware mode capture, homogeneous evolution, and size-scaling consistency. Here f_{robust} denotes the fraction of disorder-perturbed realizations that continue to satisfy the combined softness, gap, and collectiveness conditions used in the bridge diagnostics. The best-fit reduced homogeneous evolution gives

$$\frac{d \log \rho}{d \log a} \simeq -3.0001, \quad (3)$$

Metric	Twisted torus	Twisted multi ring
Flavour index	0.736	0.949
Bridge index	0.615	0.906
Overall index	0.675	0.928

TABLE II. Consolidated twisted-topology comparison from `twisted_topology_consolidated_summary.json`.

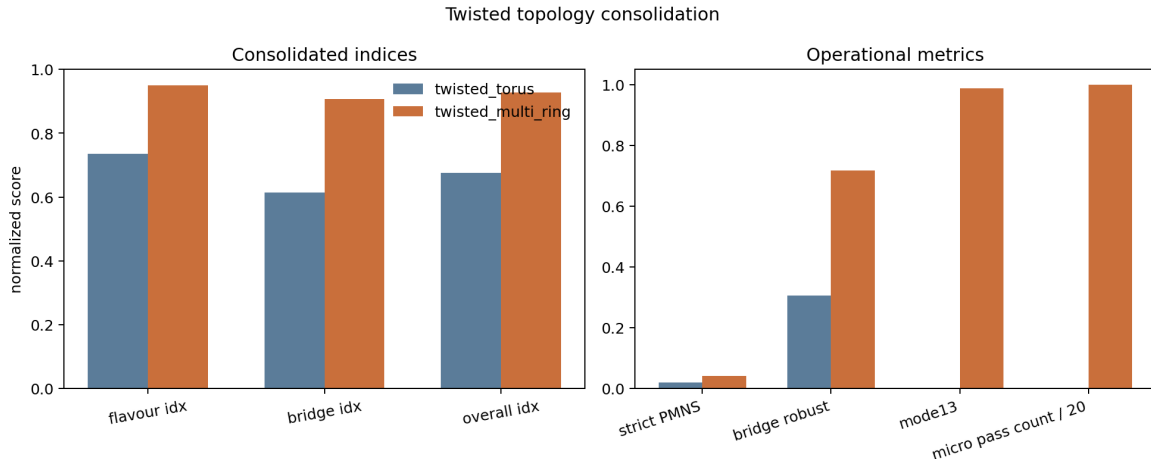


FIG. 5. Consolidated dashboard: historical twisted-torus baseline versus current twisted-multi-ring candidate.

consistent with the expected ULDM-like regime at late times. The calibrated mass lies only $|\log_{10}(m_2/10^{-22} \text{ eV})| \simeq 0.047$ decades away from the canonical ULDM target. An additional feature of the size-scaling scan is a sharp threshold: the joint fraction is zero for $N_{\text{total}} = 48$ and 72 , but rises to $\simeq 0.99$ at $N_{\text{total}} = 96$ and remains near unity above that scale. At the present stage we interpret this as evidence for a minimum-size requirement of the twisted-multi-ring topology to sustain coherent light-mode emergence, rather than as a proof of a true phase transition. This does not yet prove that Φ_2 is fully derived from first principles, but it shows that the one-field route survives a substantially stronger internal bridge test than the earlier family-level proxy stages.

VI. MICRO-CLOSURE STATUS

The one-field route would require more than an effective bridge: it would require a credible microscopic closure showing that the same $\text{CC-}\Phi_1$ network can support the effective Φ_2 sector without introducing an independent fundamental scalar.

For the earlier twisted-ring/twisted-torus-like ansatz family, repeated structured extensions eventually hit a stop rule: the programme remained structurally informative but microscopically incomplete. This remains an important caution.

The twisted-multi-ring candidate performs better. Its dedicated micro-closure attempt reaches

$$\text{verdict} = \text{micro_closure_partial_supported}, \quad (4)$$

with 20 passing records on the current grid and a best generalized projection error

$$\epsilon_{13}^{\text{proj}} \simeq 2.88 \times 10^{-2}. \quad (5)$$

This value is the minimum over the dedicated parameter grid explored in the micro-closure scan. In the present internal convention, “partial supported” means that the projection and generalized-diagonalization checks pass reproducibly while the best projection error remains at the few-percent level; a stronger “full closure” claim would require this error to be pushed to the percent level or below together with the corresponding microscopic closure conditions. This is not a theorem-level closure. It nonetheless moves the one-field interpretation from a vague aspiration to a structured and non-trivial research programme. This verdict is refined and physically interpreted in Section VII below.

Block	Primary artifact	Status label
Inverse PMNS profile	pmns_inverse_geometry_constraint_profile.json	inverse inference
PMNS/geometry scan	pmns_inverse_geometry_scan_summary.json	family ranking
Topological extension	topological_geometry_extension_summary.json	extension screening
Twisted head-to-head	twisted_topology_headtohead_summary.json	dedicated comparison
Twisted consolidation	twisted_topology_consolidated_summary.json	consolidated benchmark
Full bridge	twisted_multi_ring_full_bridge_summary.json	dedicated bridge
Micro closure	twisted_multi_ring_micro_closure_full_summary.json	partial closure
Residual diagnostic	multiring_residual_diagnostic_summary.json	operator residuals
Effective-sector test	multiring_effective_sector_diagnostic_summary.json	sector closure
Stop-rule memo	phi1_to_phi2_stop_rule_and_position.md	guardrail

TABLE III. Main script-backed artifacts used in the present Paper-VIII analysis.

VII. RESIDUAL NON-CLOSURE AND EFFECTIVE-SECTOR INTERPRETATION

The new residual diagnostics clarify why the programme gets close to closure without reaching a full microscopic theorem. The issue is not primarily the inter-ring coupling topology. For the twisted-multi-ring benchmark, the retained low-mode projector reproduces the mean source-level projection error

$$\epsilon_{13}^{\text{proj}} \simeq 4.51 \times 10^{-2}, \quad (6)$$

but the residual operator norm remains large,

$$\epsilon_{\text{op}}^{(n_{\text{keep}}=4)} \simeq 0.99996, \quad (7)$$

and the omitted spectral fraction is

$$f_{\lambda, \text{omit}} \simeq 0.9977. \quad (8)$$

Here $n_{\text{keep}} = 4$ should be read against a total mode count $N = 96$. The interpretation is direct: the low twisted sector already captures the relevant coherent source overlaps, but it does not reconstruct the full microscopic operator because most spectral weight remains in omitted higher internal modes.

The residual block structure points in the same direction. The dominant residual terms sit on the intra-ring diagonal blocks, at the level of ~ 0.57 – 0.58 , whereas the inter-ring residual blocks remain only at the level of ~ 0.028 – 0.029 . This strongly disfavors the idea that the remaining failure is mainly caused by the topology of inter-ring links. Instead, the residual is tail-dominated and largely internal to each ring.

The source-bank analysis further sharpens the picture. Global coherent twist sources such as `global_twist_k1_sin` and `global_twist_k1_cos` project almost entirely into the low twisted sector, with low-mode fractions near 0.99 and 0.98 respectively. By contrast, local contrast and twist-pair sources populate the omitted tail very strongly. This means that the current effective reduction already works for coherent global excitations, but not for the broader set of local or contrastive microscopic excitations.

This interpretation is confirmed by the enlarged-subspace scan. When the retained space is increased up to $n_{\text{keep}} = 12$, the coherent-sector mean error drops to

$$\epsilon_{\text{coh}} \simeq 8.58 \times 10^{-2}, \quad (9)$$

while local and twist-pair sectors remain much less controlled,

$$\epsilon_{\text{local}} \simeq 0.425, \quad \epsilon_{\text{twistpair}} \simeq 0.462. \quad (10)$$

The corresponding verdict is therefore not full closure but

$$\text{verdict} = \text{coherent_sector_closure_only}. \quad (11)$$

The most conservative reading is then the following: Φ_2 should not presently be interpreted as a full microscopic image of all $\text{CC-}\Phi_1$ excitations. Rather, it is better interpreted as the effective low-energy mode of a coherent global sector selected within the $\text{CC-}\Phi_1$ microstructure. This is still compatible with the one-field route, but it narrows its mathematical meaning: the route is presently supported as a coherent-sector EFT closure, not as a closure of the entire micro-operator.

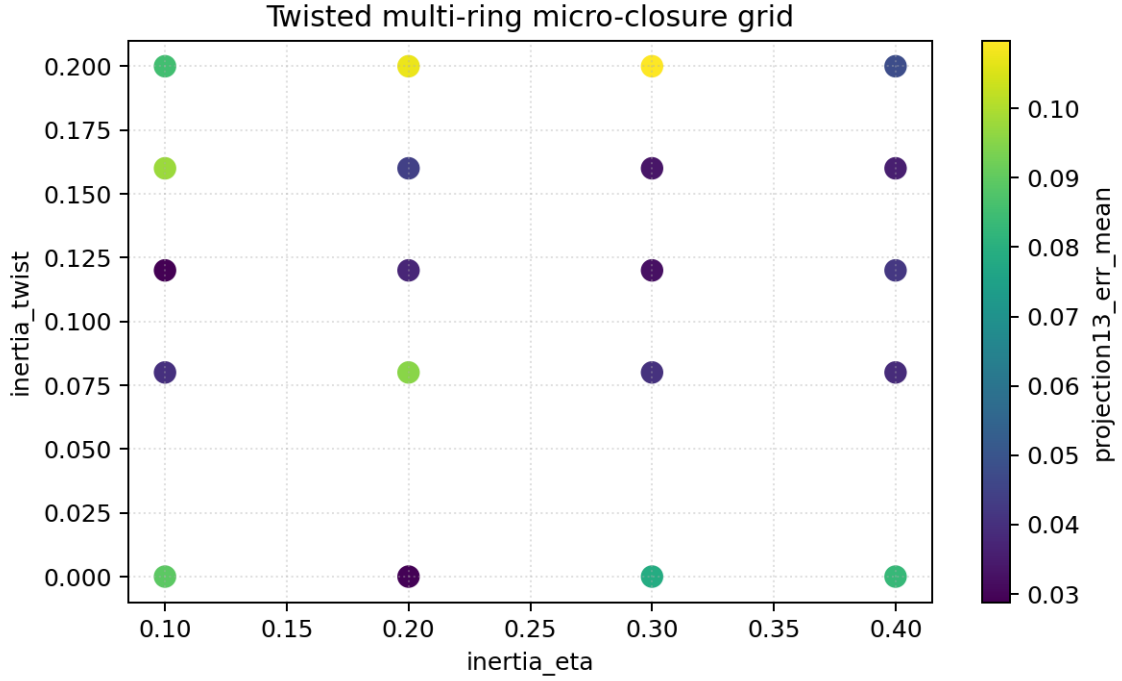


FIG. 6. Dedicated twisted-multi-ring micro-closure grid. The current status is partial support, not full closure.

VIII. INTERPRETATION

The present evidence supports three statements.

First, the geometry of the cohesive vacuum microstructure is not optional in the TCV- ϕ framework. It materially changes both flavour outputs and emergent-mode structure.

Second, the relevant class is no longer just “non-rigid geometry” in a loose sense. The data point more specifically to closed twisted topologies with several coupled channels.

Third, the one-field route is best formulated as:

$$\text{CC-}\Phi_1 \text{ microstructure} \longrightarrow \text{coherent twisted sector} \longrightarrow \text{emergent } \Phi_2 \text{ mode.}$$

This route is strongly supported structurally, but not yet closed microscopically at the full-operator level. In particular, the data suggest that the relevant ingredient for both flavour and Φ_2 emergence is not a generic deformation, but a specific class of closed twisted multi-channel topologies together with a coherent-sector projection.

IX. STATUS AND CONCLUSION

The proper current status is:

- strong support for topology-sensitive emergence;
- strong support for the twisted-multi-ring geometry as the best current joint candidate;
- reduced homogeneous ULDM-like consistency;
- partial but non-final micro-closure at the full-operator level;
- coherent-sector closure support for the effective Φ_2 mode;
- no final one-field theorem yet.

In that sense, Paper VIII should not be read as a declaration that the two-scalar EFT is obsolete. Rather, it defines the strongest current route by which the canonical EFT may admit a deeper one-field completion. The practical conclusion is therefore conservative: twisted-torus remains the historical geometry that revealed the correct topological direction, while twisted-multi-ring is now the main candidate on which future one-field and microscopic work should concentrate. The additional conclusion of the present version is more specific: the most defensible present interpretation of Φ_2 is that of a coherent-sector effective mode of $\text{CC-}\Phi_1$, not that of a complete microscopic closure of all $\text{CC-}\Phi_1$ excitations.

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